Piyush Kumar Kumawat

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PROFESSIONAL SUMMARY

Ph.D. Candidate in Chemical Engineering specializing in geothermal reservoir simulation and subsurface energy systems. Experience supporting DOE Utah FORGE and Chevron-funded research, with collaboration and internships at Idaho National Laboratory. Expertise in coupled THM modeling, fracture characterization, uncertainty quantification, and field-scale reservoir forecasting.

TECHNICAL EXPERTISE

Reservoir & Geothermal Engineering: Discrete Fracture Network (DFN) Modeling, History Matching, Fracture Connectivity Analysis, Field-Scale Reservoir Simulation

Reservoir & Simulation Software: CMG, ResFrac, Volsung, Leapfrog, MOOSE, TechLog, FEFLOW

Numerical Methods & Optimization: Finite Element Methods (FEM), Coupled Multiphysics Simulation, Multi-objective Optimization (MILP, MINLP), Robust & Stochastic Optimization, Production Planning

AI & Data-Driven Modeling: Surrogate Modeling, Explainable AI (XAI), Deep Learning for Fault Detection, Sensitivity Analysis, Uncertainty Quantification, Risk Modeling, Data-Driven Optimization, Design of Experiments

Programming & ML Frameworks: Python, MATLAB, GAMS, C, C++, PyTorch, PyMC, SHAP, DeepXDE, TensorFlow, scikit-learn

EXPERIENCE

Idaho National Laboratory May 2026 – Present; May – Aug 2025
Graduate Intern: Geothermal and Subsurface Energy Systems (2 Summers) Idaho Falls, ID

- Developed coupled Thermo-Hydro-Mechanical (THM) finite element models for EGS reservoirs.
- Developed and evaluated conceptual discrete fracture network (DFN) models, reducing computational cost by 50% while preserving fracture stability and sustainability predictions under coupled THM conditions.
- Contributed to development and validation of the fully coupled THMC simulator, FALCON.
- Executed large-scale finite element simulations to analyze stress redistribution and thermal drawdown behavior.
- Enhanced computational performance through solver optimization and mesh refinement strategies.

Energy & Geoscience Institute, University of Utah Nov 2022 – Present
Graduate Research Assistant Salt Lake City, UT

- Developed field-scale reservoir simulation models for Utah FORGE EGS to evaluate fracture connectivity, thermal drawdown, and long-term sustainability.
- Performed history matching and DFN-based fracture characterization to improve geothermal forecast reliability.
- Built surrogate modeling frameworks to accelerate reservoir forecasting and explore EGS design space.
- Conducted uncertainty quantification and sensitivity analysis for robust geothermal performance prediction.
- Optimized Borehole Thermal Energy Storage (BTES) system for Vernal City, Utah.
- Designed and deployed a pilot-scale heat exchanger for low-temperature geothermal applications, reducing drilling costs and surface footprint.

Data Analytics, Risk & Technology (DART) Laboratory, IIT Madras Mar 2022 – Jul 2022
Research Fellow AI & Risk Analytics Chennai, India

- Developed explainable AI models for fault detection via Deep Neural Networks (DNN) in industrial process systems.
- Applied data analytics and risk modeling to improve monitoring reliability and decision-making under complex multi-mode operating conditions.

Indian Institute of Technology Patna — Jul 2019 – Mar 2022
Senior Research Fellow Process Systems Engineering — India

- Developed MILP/MINLP optimization models to reduce industrial energy consumption and carbon emissions through improved production planning and resource allocation.
- Built robust optimization and machine learning frameworks to support decision-making under uncertainty for sustainable process operations.

Harsh Engineering Component Company — Jul 2018 – Jun 2019
Project Engineer — Mumbai, India

- Managed ONGC engineering projects involving hydro-testing, pigging, and pre-commissioning of oil field systems.
- Coordinated field execution, safety compliance, and client communication across multidisciplinary teams.

EDUCATION

University of Utah — 2022 – Present
Ph.D., Chemical Engineering — Salt Lake City, UT

Indian Institute of Technology Patna — 2020 – 2022
M.Tech (Research), Chemical Engineering — India

Thapar Institute of Engineering & Technology — 2014 – 2018
Bachelor of Engineering, Chemical Engineering — India

PUBLICATIONS and ACHIEVEMENTS

- **8+ peer-reviewed journal publications** in Renewable Energy, Geothermics, Chemical Engineering Science, Energy Storage and Savings, Colloids and Surfaces A, Clean Technologies and Environmental Policy, Case studies in Thermal Engineering
- **16+ international conference publications and presentations**: ARMA Geomechanics Symposium (2025,2026); Stanford Geothermal Workshop (2025, 2026); Geothermal Rising Conference(2025), SPE Energy Transition Symposium (2025), International Meeting for Applied Geoscience and Energy (IMAGE-2025), AIChE Annual Meet (2022, 2024, 2025)
- **Invited Talks**: Longetivity of Utah FORGE (June 2024; Utah Geological Survey, Department of Natural Resources, Salt Lake City, UT), How to Build Career in Research? (Jan 2023, Thapar Institute of Engineering and Technology)
- **Book Chapter**: Optimization for energy systems and supply chains, CRC Press

AWARDS

- Kistler Graduate Fellowship University of Utah
- Travel Award Geothermal Rising Conference, Nevada, Reno (2025)
- SERB Master's Fellowship - Government of India
- Academic Scholarship for Undergraduate Studies – Thapar Institute of Engineering and Technology, Patiala

Leadership & Service

- **Teaching Assistant**, UoU: Chemical Process Safety (CHEN 3253); Geothermal Engineering (CHEN 6960)
- **Student Volunteer**, Geothermal Rising Conference, Reno, NV (2025): Registration and Session Chair Assistant
- **Social Media & Outreach Manager** (2024-2026), ARMA Student Chapter, University of Utah
- **Professional Memberships**: ARMA, IMAGE, SPE, Geothermal Rising (GRC), AIChE